

Work Experience

Teaching/Advising Assistant	University of California Davis Davis, CA	Sep 2024–Present
- Working as a Teaching Assistant for ECS 174 (Computer Vision) and ECS 36B (C++, OOP, Software Development) - Developed automated grading scripts and curriculum for ECS 174 and 36B, facilitating technical discussions for 100+ students		
SWE Intern/Part Time	California Department of Transportation Department of Construction Sacramento, CA	Mar 2023–Feb 2026
- Worked as the full stack developer intern as a part of a 8 person team with responsibilities of getting feedback from users and maintaining/developing new features according to their feedback and needs - Built and maintained 2 full-stack React/Flask/MySQL applications (using PHP, HTML, CSS, JS, Python, Angular) used by 1,500+ users, saving 50,000+ reports of paper and thousands of labor hours		
Research Assistant	University of California Davis CompFab Lab/C^2 Davis, CA	Fall 2022–Present
- Engineered C++ simulations using the eigen library to model material folding, resulting in more accurate simulations - Fine tuning large foundational models to perform well at specific video classification tasks using sota models - Developed a Python-based ML pipeline (TensorFlow, Keras) using SVMs to classify FMRI neuroimaging data with above-random accuracy		

Education

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| • M.S. Computer Science , University of California, Davis GPA: 3.77 | Sep 2024–Present |
| • B.S. Computer Science, Minor Math , University of California, Davis | Sep 2020–June 2024 |

Technologies

- Languages: C, C++, Java, Javascript, Python, PHP, HTML, CSS, X86-64, RISC-V
- Technologies: MySQL, Git, Linux, Windows, Bash, Scripting, Cursor, Copilot
- Other: Computer Architecture, Networks, OS, Data Structures/Algorithms, AI/ML, Computer Networks

Certifications

- Amazon Web Services Cloud Practitioner
- Oracle AI Vector Search Professional

Projects

- Interpretable AI Toxicity Detector (Winter 2025)**: Developed an interpretable NSFW text moderation pipeline by fine-tuning RoBERTa (87% validation accuracy) and integrating RISE perturbation-based saliency with a post-hoc Concept Bottleneck Model to provide human-readable rationales for toxicity detection
- Differential Testing of ML Libraries (Winter 2025)**: Engineered an automated testbed to profile CPU performance and verify numerical stability across PyTorch and TensorFlow when handling edge cases like NaN, Inf, and extreme tensor values.
- Predicting California Housing Prices (Spring 2022)**: Developed a California housing price predictor in a team of 5 students using Scikit-Learn regression models and a Flask/JS web demo
- AI/ML/CV Projects (Spring 2022 and 2023)**: Coded independently to climb Mount Saint Helens using A star, implement alpha beta pruning with mini-max algorithm to play connect four autonomously, and use Deep Q learning (tensorflow) to create a model to play pong. Stitched images together using novel CV techniques like RANSAC. Classified the CIFAR dataset of images using a neural network (Pytorch). Used OCI to deploy a chatbot.
- Operating Systems Projects (Winter 2022)**: Worked as a team of two to integrate syscalls into code to emulate a shell, created an API of user level threads with preemption, and made a FAT file system (every project coded in C).